

Operator Discovery Report: structure_aware_pruner_midlim it

- Operator type: candidate_pruner.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Adapt local-search candidate-list size to instance structure using moderate stagnation and trap-aware pruning. Keeps enough moves for transfer, while reducing noise on clearer clustered/bottleneck instances.
- Novelty classification: descriptor_adaptive_pruner.
- Rediscovery signature: candidate_pruner:18:-6:3:6.
- Surviving candidate: False.
- Problem class where it helps: none confirmed.

Pseudocode

- Start from a base candidate-list size.
- Expand or shrink the list using instance descriptors such as size, trap score, grid-likeness, clustering, and current stagnation.
- Run the downstream local search only on the resulting pruned neighborhood.

Complexity

- Non-empty lines: 16.
- Complexity score: 0.52.
- Fingerprint: 2d5de5d2aa330db6.

Why It Should Help

- The operator adapts neighborhood size to instance structure, which can trade off search breadth and runtime in a reusable way.

Failure Cases

- Does not transplant cleanly into clustered_local_search.
- No validation family shows a clear improvement over the host baseline.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.
- Runtime inflation is high at 0.272308.

Transplant Results

- nearest_neighbor_2opt gap delta 0.000609, runtime delta 35.950571, distance-eval delta 1907.857143.
- cheapest_insertion_2opt gap delta 0.0, runtime delta 1150.2946, distance-eval delta 1007.428571.
- random_restart_2opt gap delta 0.001067, runtime delta 21.557186, distance-eval delta 915.857143.
- sparse_three_opt gap delta -0.000253, runtime delta 51.439114, distance-eval delta 3976.0.
- clustered_local_search gap delta 0.032473, runtime delta 228.796757, distance-eval delta 6803.142857.

Instance-Family Results

- heldout_tsplib operator gap 0.235667 vs baseline 0.235058.
- clustered_tsp operator gap 0.110853 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: 0.000224.
- Runtime inflation: 0.272308.
- Distance-evaluation inflation: 0.19862.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator -0.000224.
- constant_pruner mean gap 0.121542 and delta vs full operator 0.0.
- wide_pruner mean gap 0.121542 and delta vs full operator 0.0.